E-Commerce Conversion Prediction

Summer Analytics 2026 - Week 2 Hackathon | One-Page Methodology Report

1. Problem Statement

Binary classification task to predict whether a user will convert (make a purchase) based on anonymized user demographics, browsing behavior, purchase history, device info, and session-level activity. Evaluation metric: F1 Score on private test set.

2. Dataset Overview

Dataset	Rows	Features	Target
train.csv	10,000	13	Converted (0/1)
public_test.csv	3,000	13	Converted (0/1)
private_test.csv	3,000	12	Unknown

Class imbalance: 69.1% non-converted (0) vs 30.9% converted (1). Missing values present in Age (14.8%), Income (9.8%), Time_On_Site (18.5%).

3. Preprocessing & Feature Engineering

Missing Value Handling: Median imputation for Age, Income, and Time_On_Site using sklearn SimpleImputer fitted only on training data.

Categorical Encoding: Label Encoding for Device_Type (Mobile/Desktop/Tablet) and Traffic_Source (5 categories).

New Features Created (7):

Feature	Formula	Rationale
Pages_per_Product	Pages_Viewed / (Products_Viewed+1)	Browsing depth per product
Time_per_Page	Time_On_Site / (Pages_Viewed+1)	Engagement intensity
Engagement_Score	Pages x Time / (Products+1)	Overall session quality
Purchase_Discount_Interaction	Prev_Purchases x Discount_Seen	Discount sensitivity
High_Value_User	Income > median & Purchases > 2	Premium user flag
Log_Income	log1p(Income)	Normalize skewed income
Age_Group	Bins: <25, 25-35, 35-50, 50+	Generational segments

4. Model Architecture - Weighted Ensemble

Three complementary models were trained on the full training set and their probability outputs combined using a weighted average (RF: 35%, GB: 35%, ET: 30%). This reduces variance and improves generalization compared to any single model.

Model	Key Parameters	Weight	Rationale
Random Forest	n=300, depth=10, balanced	35%	Robust, handles imbalance
Gradient Boosting	n=200, depth=5, lr=0.1	35%	Best at capturing patterns
Extra Trees	n=300, depth=10, balanced	30%	High variance reduction

5. Threshold Optimization

Default threshold of 0.5 is suboptimal for imbalanced datasets. The ensemble probability output was evaluated across thresholds from 0.20 to 0.80 on the public test set. **Optimal threshold: 0.32**, achieving **Public F1 Score: 0.5499**. This lower threshold increases recall for the minority class (converted=1), which is critical for maximizing F1 in imbalanced scenarios.

6. Top Predictive Features

Based on Random Forest feature importances, the most influential features were: **Campaign_Code, Engagement_Score, Time_On_Site, Income, Log_Income, Age, Pages_Viewed**, and **Previous_Purchases**. This confirms that session engagement metrics and user demographics are the strongest predictors of conversion.

7. Results Summary

Metric	Value
Optimal Classification Threshold	0.32
Public Test F1 Score	0.5499
Total Features Used	19 (12 original + 7 engineered)
Cross-validation Strategy	5-Fold Stratified
Private Test Predictions: 0 / 1	1240 / 1760